20 Word Problems: Questions

Rational Numbers Word Problems (show all work)

1. HIKER: A hiker starts at -60 feet (below sea level). She climbs 2/3 of that distance toward sea level, then descends 12.5 feet. What is her final elevation?

2. SUBMARINE: A submarine at -240 feet rises 3/8 of its depth, then dives 45.5 feet. What is its new elevation?

3. WEATHER: The temperature is -12.4 F. It rises by 1/4 of that amount, then drops 5.75 F. What is the final temperature?

4. MINE ELEVATOR: An elevator is at -350 feet. It rises 4/5 of the way to the surface, then goes down 22.5 feet. Where is it now?

5. BANK: An account balance is -$84.60. A deposit covers half of what is owed, then $17.25 is spent. What is the balance?

6. FREE DIVER: A diver at -36 feet rises 5/6 of the way up, then descends 9.25 feet. What is her final depth?

7. CAVE: An explorer at -128 feet climbs 3/4 of the way up, then goes down 14.6 feet. Where is the explorer?

8. STOCKS: A stock changes by -2.4, +1 3/4, -0.85, and +0.5 dollars over four days. What is the average daily change?

9. TEMPERATURE: Daily changes from the average are -3.5, -1 1/4, +2.75, -4, and +0.5 degrees. What is the average change?

10. WEIGHT: A dog's weight changes by -1.6, +0.4, -2 1/5, and -0.6 pounds over four weeks. What is the average weekly change?

11. LAKE: A lake's level changes by +0.75, -1 1/2, -2.25, +0.5, -1.75, and -0.25 inches over six months. What is the average monthly change?

12. BUSINESS: Quarterly profits are -3.2, +5.6, -1 3/5, and +2.4 million dollars. What is the average quarterly profit?

13. SPENDING: Over five days an account changes by -$12.50, +$30.25, -$8.75, -$4.40, and +$15. What is the average daily change?

14. BAKING: One batch needs 3 1/4 cups of sugar. With 20 cups, how many full batches can you make, and how much is left?

15. BOWS: Each bow uses 1 3/8 yards of ribbon. With 15 yards, how many bows can you make, and how much ribbon is left?

16. TRAIL MIX: How many 2.25-pound bags can be filled from 30 pounds, and how much is left?

17. JUICE: How many 0.75-liter bottles can be filled from 8.6 liters, and how much is left?

18. SHELVES: A 12-foot board is cut into shelves 2 5/6 feet long. How many shelves, and how much board is left?

19. COSTUMES: Each costume needs 1 3/4 yards of fabric. With 9 1/2 yards, how many costumes, and how much fabric is left?

20. PIZZA: There are 4 2/3 pizzas, and each person eats 5/6 of a pizza. How many people can be fully served, and how much pizza is left?